Go2 SDK Development Guide

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Robot Dog Application

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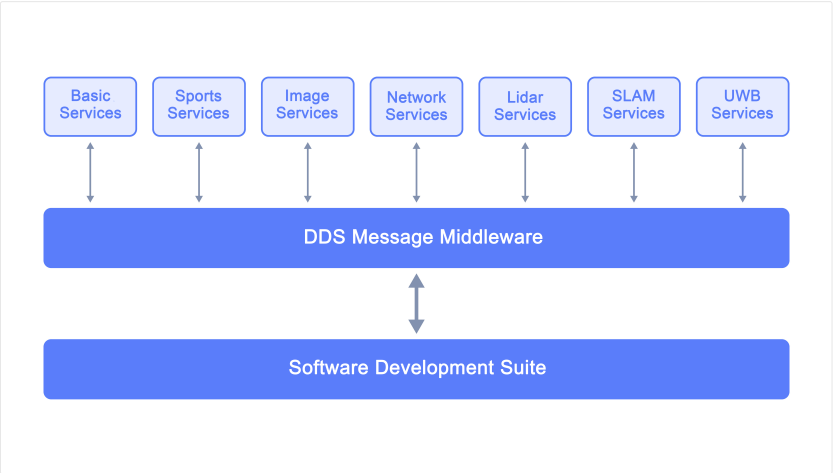

The machine dog provides the following services through DDS message middleware:

- Basic services

Obtain real-time running data of the robotic dog, including batteries, motors, IMUs, etc. Simultaneously providing a low-level control interface (directly driving the motor to control gait)

- Sports Services

From a macro perspective, control the movement of the robotic dog, with functions such as switching gait, controlling posture and speed, performing stunts or dancing.

- Image Services

Photography, image streaming, and other functions

- Network Services

View 4G IoT network card information, configure Wi Fi, AP, and STA modes.

- Radar Services

Obtain point cloud data and IMU data, construct altitude maps and 3D voxel maps.

- SLAM Services

It is divided into three parts: mapping, positioning, and navigation. Mapping is used to generate the current environment map for repositioning, determine the robot's pose in the current environment, and the navigation module automatically plans the path to the target point based on the positioning results and the deviation from the target point.

- UWB Service

Based on the self-developed vector positioning module, it can achieve centimeter level positioning in the plane of the fuselage. The tag is equipped with a 3-axis accelerometer and a 3-axis gyroscope IMU, which can provide the attitude of the tag. Mainly used as a companion function for Go2 integration, it can also be used for user secondary development.

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